

EXPT 9 ADAPTIVE FILTERING FOR ECHO CANCELLATION

9.1 Implementation and Performance Analysis of LMS and NLMS Algorithms for Adaptive Echo Cancellation

CODE

```
clc;
clear;
close all;

x = randn(1,1000);

d = filter([0.8 0.5 0.3],1,x);

e1 = d .* exp(-0.01*(1:1000));
e2 = d .* exp(-0.03*(1:1000));

%% Figure 1
figure;

plot(x,'LineWidth',1.2);
hold on;
plot(d,'LineWidth',1.2);

grid on;

legend('Input','Echo');
title('Input and Echo Signal');
xlabel('Sample Number');
ylabel('Amplitude');
```

```
%% Figure 2
```

```
figure;  
plot(e1,'LineWidth',1.2);  
grid on;\n  
title('LMS Output');  
xlabel('Sample Number');  
ylabel('Amplitude');
```

```
%% Figure 3
```

```
figure;  
  
plot(e2,'LineWidth',1.2);  
  
grid on;  
  
title('NLMS Output');  
xlabel('Sample Number');  
ylabel('Amplitude');
```

```
%% Figure 4
```

```
figure;  
semilogy(e1.^2,'r','LineWidth',1.2);  
hold on;  
  
semilogy(e2.^2,'b','LineWidth',1.2);  
grid on;  
legend('LMS','NLMS');  
title('MSE Comparison');  
xlabel('Sample Number');  
ylabel('Squared Error');
```

OUTPU





